MYSQL ASSIGNMENT2

-QUERYING DATA

```
CREATE DATABASE Employee;

USE Employee;


-- 1.TABLE CREATION FOR DEPARTMENTS,EMPLOYEE,LOCATION:


# 1.Department Table:

CREATE TABLE departments(
        department_id INT PRIMARY KEY,
        department_name VARCHAR(100) NOT NULL UNIQUE
);


# 2. Location Table:

CREATE TABLE location(
        location_id INT AUTO_INCREMENT PRIMARY KEY,
        location VARCHAR(30) NOT NULL UNIQUE
);
```

3. Employees Table:

```
CREATE TABLE employees (
        employee_id INT PRIMARY KEY,
        employee_name VARCHAR (50) NOT NULL,
        gender ENUM('M','F'),
        age INT CHECK(age>=18),
        hire_date DATE DEFAULT(current_date()),
        designation VARCHAR (100),
        department_id INT,
        location_id INT,
        salary DECIMAL (10,2),
        FOREIGN KEY(department_id)
        REFERENCES departments(department_id),
        FOREIGN KEY(location_id)
        REFERENCES location(location_id)
);



SELECT * FROM departments;
SELECT * FROM location;
SELECT * FROM employees;
```

```
INSERT INTO departments (department_id, department_name) VALUES
(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
(7, 'Finance'),
(8, 'Design'),
(9, 'Research and Development'),
(10, 'Customer Support'),
(11, 'Business Development'),
(12, 'IT'),
(13, 'Operations');

INSERT INTO location (location) VALUES
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');
```

```
INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary) VALUES

(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),

(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),

(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),

(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),

(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),

(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),

(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),

(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),

(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),

(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),

(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),

(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),

(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),

(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),

(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),

(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),

(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),

(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),
```

(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),

(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),

(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),

(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),

(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),

(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),

(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),

(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),

(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),

(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),

(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),

(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);

-- 1.Distinct Values:

```
SELECT DISTINCT salary FROM employees;
```

-- 2.Alias (AS):

```
SELECT *,age AS Employee_Age,salary AS Employee_Salary FROM employees;
```

-- 3.Where Clause & Operators:

employees with a salary greater than ₹50000 and hired before 2016-01-01

```
SELECT * FROM  employees WHERE salary>50000 AND hire_date<'2016-01-01';
```

employee whose designation is missing and fill it with "Data Scientist"

```
UPDATE employees SET designation = "Data Scientist" WHERE designation IS NULL AND employee_id>0;

SELECT * FROM employees;
```

-- Sorting and Grouping Data: --

-- 1.ORDER BY:

```
SELECT * FROM employees ORDER BY department_id ASC,salary DESC;
```

-- 2.LIMIT:

```
SELECT * FROM employees WHERE YEAR(hire_date)=2018 LIMIT 5;
```

-- 3.Aggregate Functions:

Sum of all salaries in the Finance department

```
SELECT sum(e.salary) AS Total_salary FROM employees e JOIN departments d ON e.department_id=d.department_id WHERE d.department_name="Finance";
```

Minimum age among all employees

```
SELECT min(age) AS Minimum_age FROM employees;
```

-- 4.GROUP BY:

Maximum salary for each location

```
SELECT l.location,max(e.salary) AS maximum_salary FROM employees e JOIN location l ON e.location_id=l.location_id GROUP BY l.location_id,l.location;
```

Average salary for each designation containing the word 'Analyst'

```
SELECT designation,AVG(salary) AS average_salary FROM employees WHERE designation LIKE '%Analyst%' GROUP BY designation;
```

-- 5.HAVING:

Departments with less than 3 employees

```
SELECT d.department_name,COUNT(e.employee_id) AS Employee_count FROM departments d JOIN employees e ON d.department_id=e.department_id GROUP BY d.department_id,d.department_name HAVING COUNT(e.employee_id)<3;
```

locations with female employees whose average age is below 30.

```
SELECT l.location,AVG(age) AS Average_age FROM employees e JOIN location l ON e.location_id=l.location_id WHERE e.gender='F' GROUP BY l.location HAVING AVG(e.age)<30;
```

-- JOINS: --

1.Inner Join:

```
SELECT e.employee_name,e.designation,d.department_name FROM
employees e INNER JOIN departments d ON
e.department_id=d.department_id;
```

2.Left Join:

```
SELECT d.department_name,count(e.employee_id) AS Total_Employees FROM
departments d LEFT JOIN employees e ON d.department_id=e.department_id
GROUP BY d.department_id,d.department_name;
```

3.Right Join:

```
SELECT l.location,e.employee_name FROM employees e RIGHT JOIN location l
ON e.location_id=l.location_id;
```